# Chapter 19

A Macroeconomic Theory of Open Economy

# Introduction

- Now we will look to understand how different markets work together in equilibrium
- In previous chapter we studied concepts and definitions
- In this chapter, we will see how the market for loanable funds and market for Foreign exchange…
- Work together to determine the economy's trade balance and exchange rate

# Supply and Demand for Loanable Funds

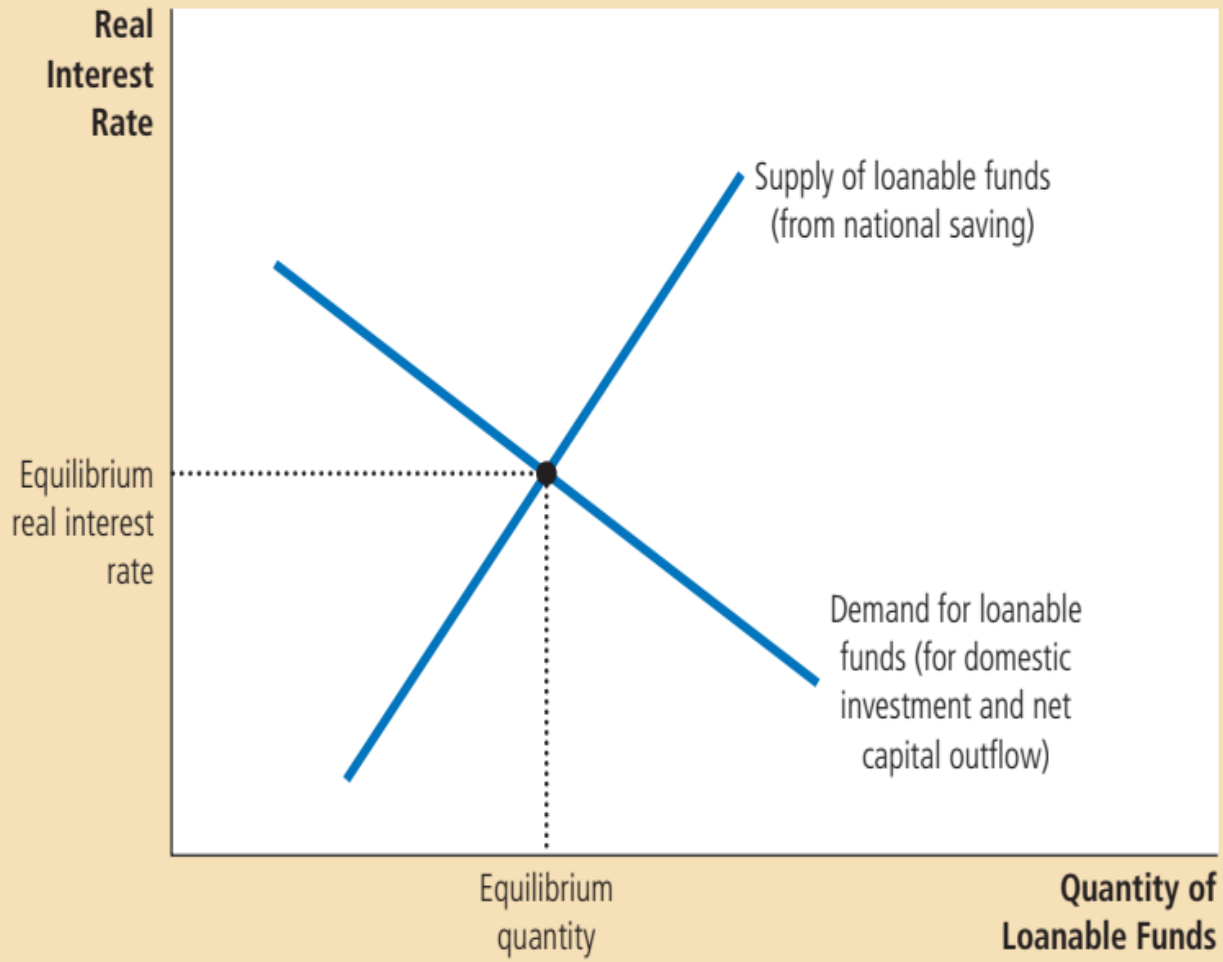
- Earlier, S = I
- In Open Economy case, S = I + NCO
- So Supply of loanable funds comes from Savings, demands for loanable funds from Invt + NCO
- If NCO<0 => capital inflow => reduced demand for domestic funds
- If NCO>0 => capital outflow => greater demand or domestic funds abroad

- Supply and Demand depend on the real interest rate
- Real IR affects NCO: Higher RIR in one country encourages flow of funds to that country

Figure 1

**The Market for Loanable Funds**

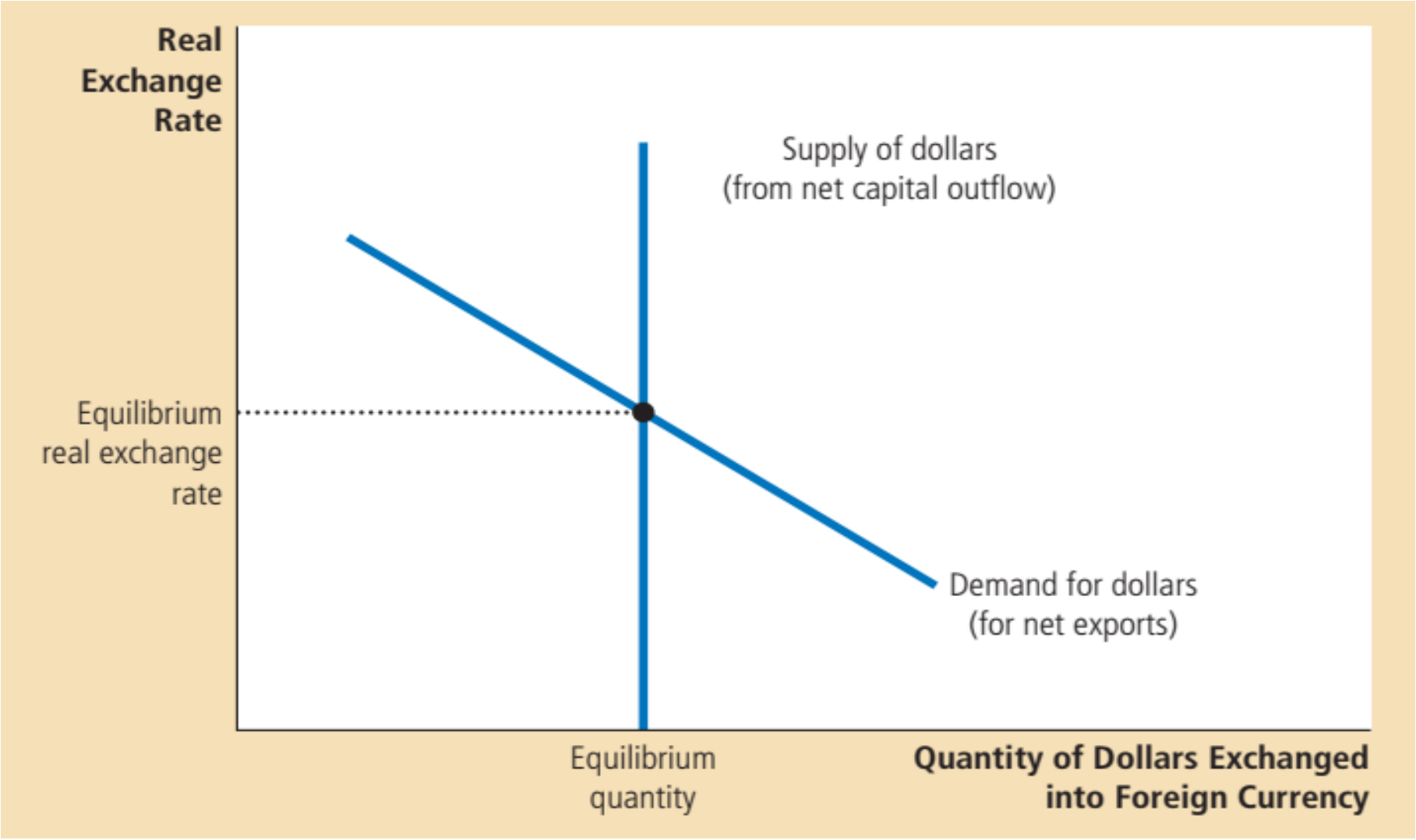
The interest rate in an open economy, as in a closed economy, is determined by the supply and demand for loanable funds. National saving is the source of the supply of loanable funds. Domestic investment and net capital outflow are the sources of the demand for loanable funds. At the equilibrium interest rate, the amount that people want to save exactly balances the amount that people want to borrow for the purpose of buying domestic capital and foreign assets.



# Market for Foreign Currency Exchange

- Key Equation in this market is NX=NCO.
- If NX>0 => NCO>0: domestic residents are buying foreign assets
- NCO represents quantity of dollars supplied for the purpose of buying foreign assets (US residents supplying dollars to buy foreign currency which are further used to buy foreign assets)
- NX represents demand for dollars: exporters to US (domestic) demand dollars to buy American goods and services

- What is the price for demand and supply for foreign exchange (key determinant of net exports)?
- RER is relative price of foreign goods in terms of domestic goods
- US dollar (domestic) appreciates, exports decrease, less demand for US dollars by foreign buyers
- US $ depreciates, more demand for dollars


Real Exchange Rate
Supply of dollars (from net capital outflow)
Equilibrium real exchange rate
Demand for dollars (for net exports)
Equilibrium quantity
Quantity of Dollars Exchanged into Foreign Currency

- Why is supply curve vertical? NCO does not depend on RER
- Why?

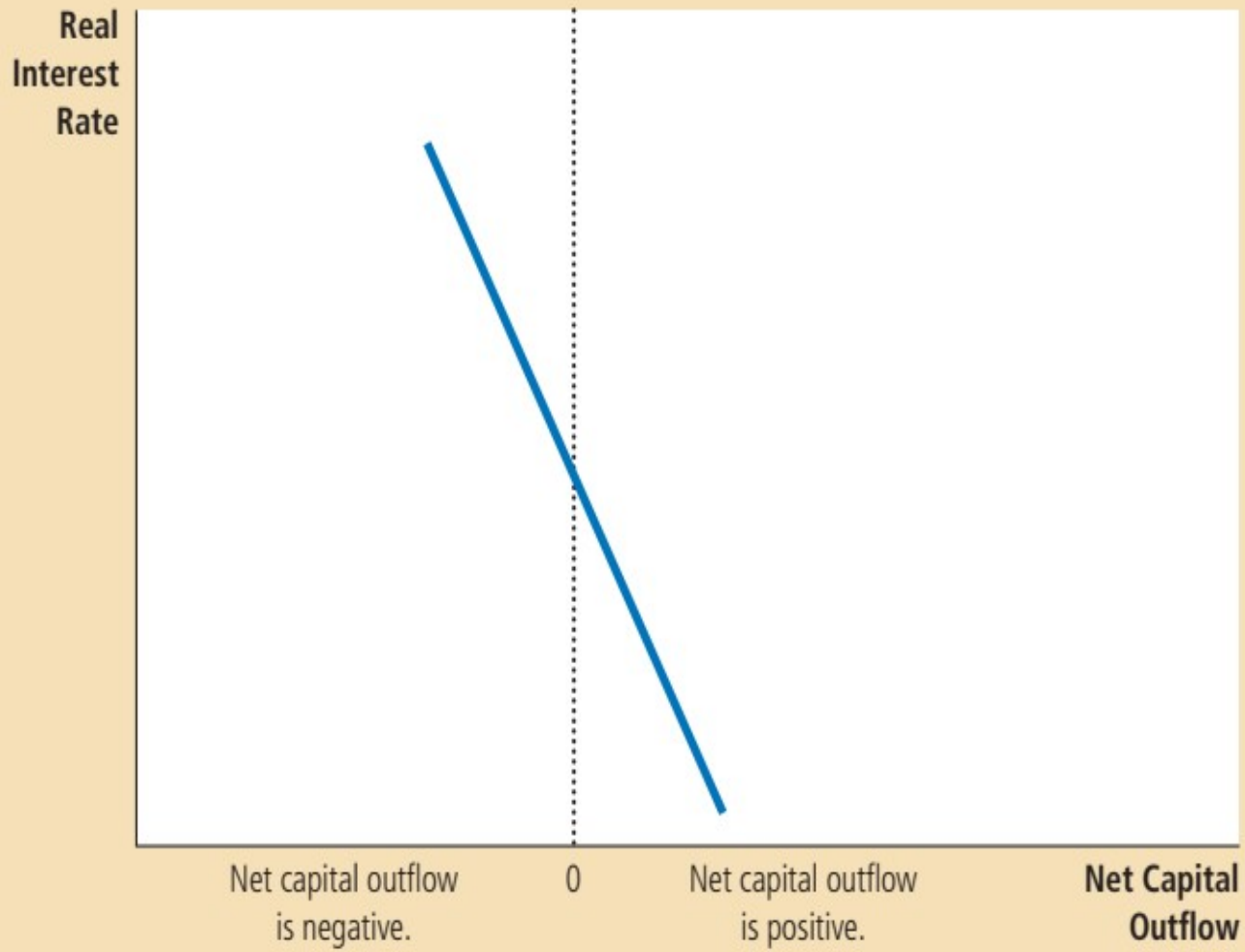
# Equilibrium in the Open Economy

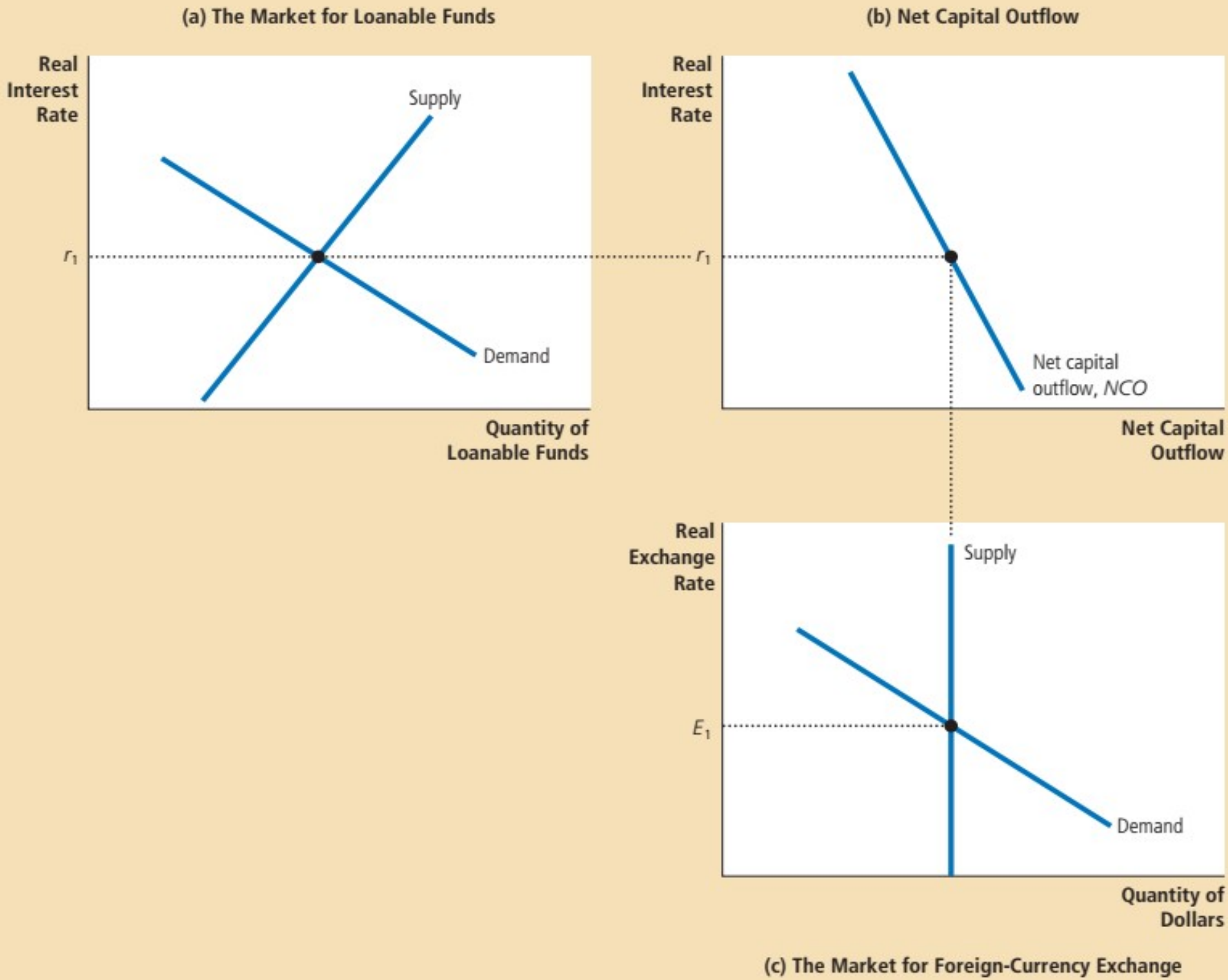
- So we have two markets, determined each by two equations
- First: S = I + NCO
- Second: NX = NCO;
- NCO is part of demand in market for loanable funds
- And supply in market for foreign exchange
- The key determinant of NCO is Real Interest Rate

Figure 3

**How Net Capital Outflow Depends on the Interest Rate**

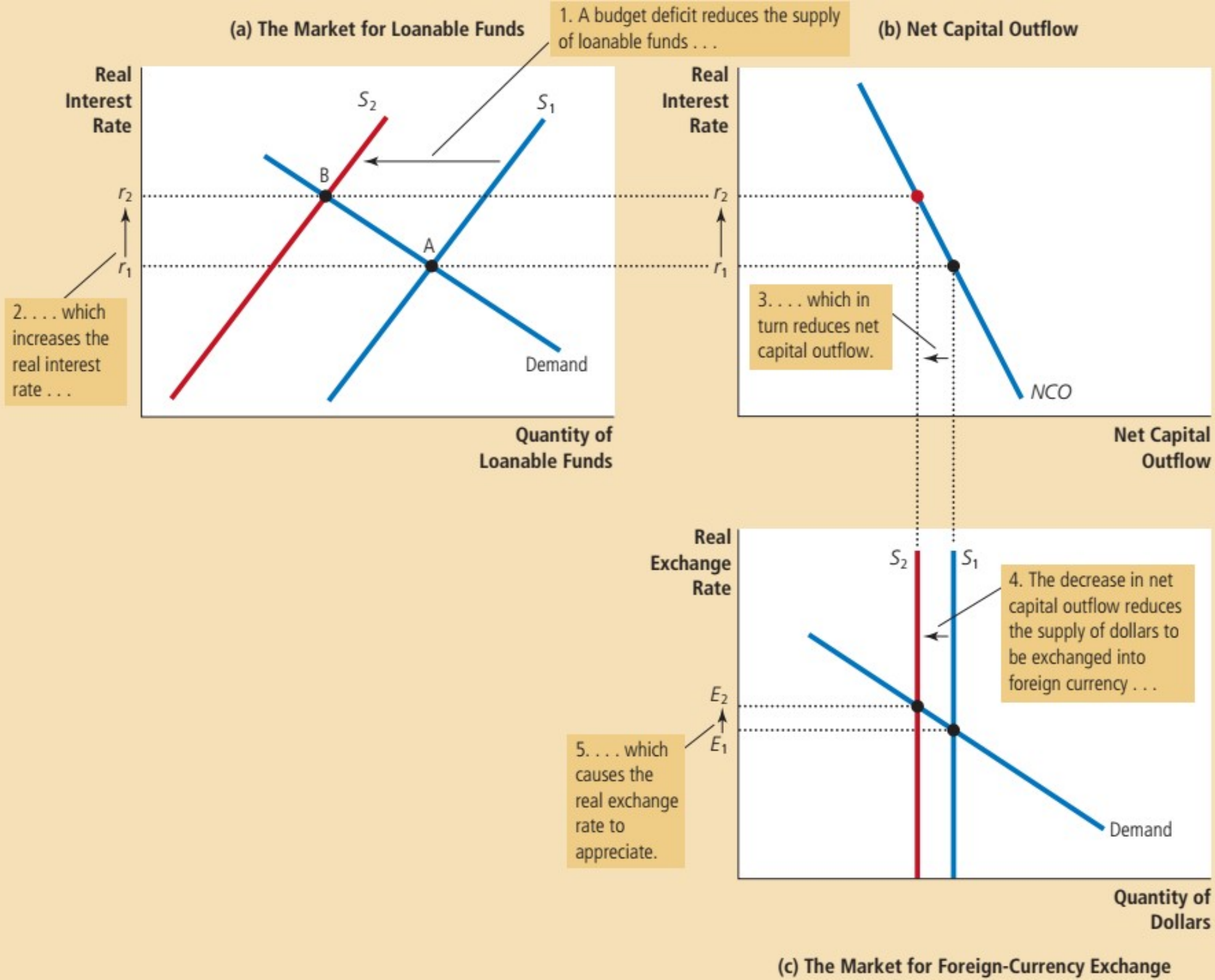
Because a higher domestic real interest rate makes domestic assets more attractive, it reduces net capital outflow. Note the position of zero on the horizontal axis: Net capital outflow can be positive or negative. A negative value of net capital outflow means that the economy is experiencing a net inflow of capital.




(a) The Market for Loanable Funds
Real Interest Rate
Supply
$r_1$
Demand
Quantity of Loanable Funds
(b) Net Capital Outflow
Real Interest Rate
$r_1$
Net capital outflow, *NCO*
Net Capital Outflow
Real Exchange Rate
Supply
$E_1$
Demand
Quantity of Dollars
(c) The Market for Foreign-Currency Exchange

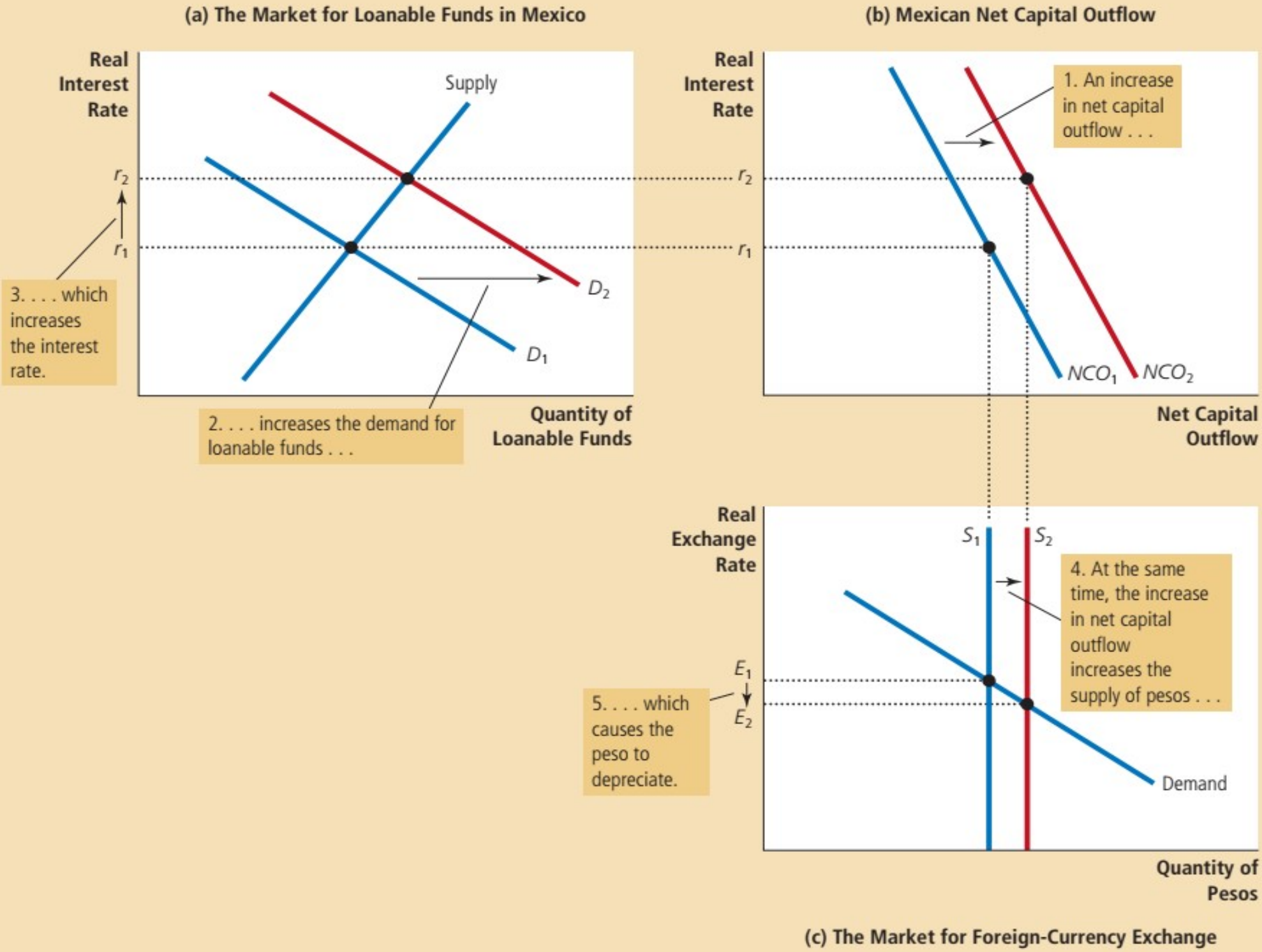
# What affects the equilibrium?

- Government budget deficits => negative public savings => reduces national savings => leftward shift of supply of loanable funds => shortage of supply => increase in real interest rate
- NCO ***moves*** left
- NCO shifts left => supply curve of foreign exchange shifts leftwards => exchange rate increases, net exports fall.


(a) The Market for Loanable Funds
1. A budget deficit reduces the supply of loanable funds . . .
Real Interest Rate
$S_2$
$S_1$
B
A
$r_2$
$r_1$
2. . . . which increases the real interest rate . . .
Demand
Quantity of Loanable Funds
(b) Net Capital Outflow
Real Interest Rate
$r_2$
$r_1$
3. . . . which in turn reduces net capital outflow.
NCO
Net Capital Outflow
Real Exchange Rate
$S_2$
$S_1$
4. The decrease in net capital outflow reduces the supply of dollars to be exchanged into foreign currency . . .
$E_2$
$E_1$
5. . . . which causes the real exchange rate to appreciate.
Demand
Quantity of Dollars
(c) The Market for Foreign-Currency Exchange

# Political Instability and Capital Flight

- Political instability causes capital to move out, to ‘safe’ havens.
- Model applied with Mexico as domestic and US as foreign country.
- NCO increases, as Mexicans invest money abroad instead of at home, and other inflows reduce => demand curve for loanable funds shifts right
- Exogenous increase in NCO: shift or movement?


(a) The Market for Loanable Funds in Mexico
Real Interest Rate
Supply
$r_2$
$r_1$
$D_2$
$D_1$
3. . . . which increases the interest rate.
2. . . . increases the demand for loanable funds . . .
Quantity of Loanable Funds
(b) Mexican Net Capital Outflow
Real Interest Rate
1. An increase in net capital outflow . . .
$r_2$
$r_1$
$NCO_1$
$NCO_2$
Net Capital Outflow
Real Exchange Rate
$S_1$
$S_2$
4. At the same time, the increase in net capital outflow increases the supply of pesos . . .
$E_1$
$E_2$
5. . . . which causes the peso to depreciate.
Demand
Quantity of Pesos
(c) The Market for Foreign-Currency Exchange